

2.3 LSTMs and GRUs

Gates keep a near-1 path through time. LSTM has a cell; GRU does not.

These notes match the lecture slides. Use **(slides)** on the course hub for the deck.

Gates are learned valves. They let a recurrent cell **keep** a bit of memory for many steps instead of overwriting it every time. LSTM and GRU are the two you will meet. They do not remove vanishing gradients; they make a near-1 path available.

1. LSTM: a highway plus three gates

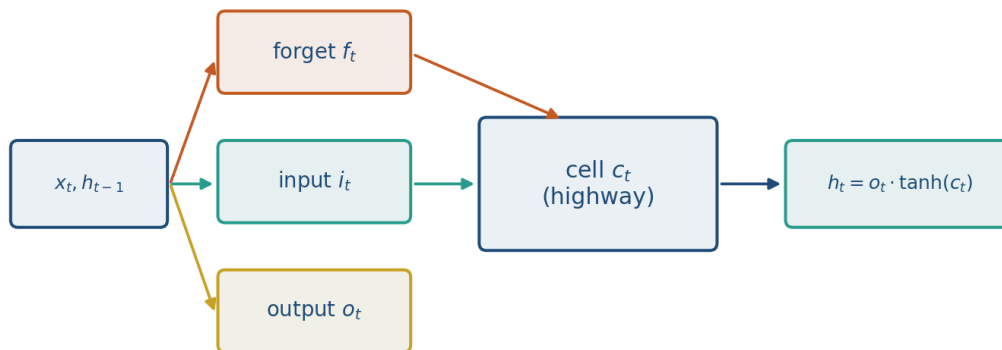
Hochreiter and Schmidhuber (1997). A **cell state** c_t moves down the chain with mostly multiplies and adds, not a stack of tanhs. Three sigmoid gates, each in $(0, 1)$:

Gate	Role
Forget f_t	How much of c_{t-1} to keep
Input i_t	How much new candidate to write
Output o_t	How much of $\tanh(c_t)$ becomes h_t

$$c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t, \quad h_t = o_t \odot \tanh(c_t).$$

If $f_t \approx 1$ and $i_t \approx 0$, the cell copies. That is the long-range path.

LSTM: a cell highway plus learned gates



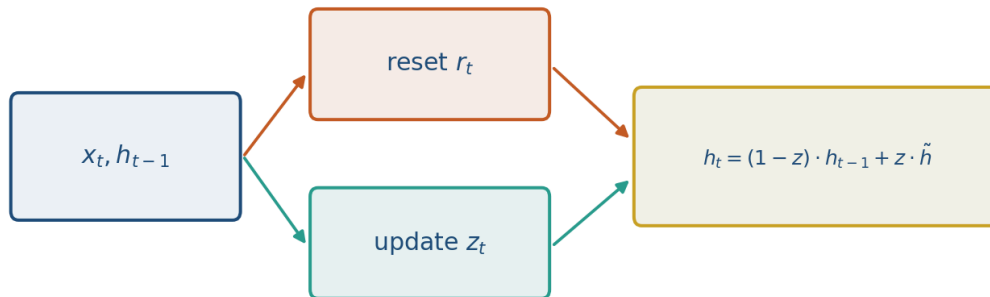
2. GRU: two gates, no extra cell

Cho et al. (2014). Reset r_t and update z_t . The hidden state *is* the memory.

$$h_t = (1 - z_t) \odot h_{t-1} + z_t \odot \tilde{h}_t.$$

Fewer parameters, often similar quality on medium sequences. Use LSTM when you want an explicit cell; use GRU when you want a smaller default.

GRU: two gates, no separate cell state



3. What gates do not buy you

The loop is still sequential. You still cannot fill a GPU the way a transformer can. Long context is better than vanilla, worse than attention. For this course, gates are the reason we can talk about “memory” before Week 3, and a baseline in speech/time-series projects.

Bidirectional LSTM: run one net forward and one backward, concatenate. Fine for classification. Illegal for next-token generation (you would peek at the future).

4. Practice

1. If the forget gate is stuck at 0, what happens to c_t ?
2. Why is a bidirectional LSTM the wrong cell for a language model that must emit token $t + 1$?
3. Name one reason you might still pick a GRU over a transformer in a project.